

Week 1: Tilings

Facilitator guide for the three adults

One hour · ten children · three tables

Contents

1 The session at a glance	1	6.3 Puzzle 3: two boards with 14 squares (page 3)	10
2 Materials and preparation	2	6.4 Puzzle 4: the secret coloring (page 4) . . .	10
3 Minutes 0–5: free handling	3	6.5 Puzzles 5 and 6: the big board (page 5) . .	11
4 Minutes 5–12: the launch (organizer)	3	6.6 Puzzles 7 and 8: extensions (pages 6–7) .	11
5 Table K–1: script for the parent	4	7 Table 4–5: the L-tromino and the board that doubles	13
5.1 Puzzle 1: fill the hexagon (page 1)	4	7.1 Puzzles 1–3: the 4×4 with a hole (pages 1–2)	13
5.2 Puzzle 2: the small triangle (page 2) . . .	5	7.2 Puzzles 4 and 5: the trick and the counting check (page 3)	14
5.3 Puzzle 3: the big triangle (page 3)	6	7.3 Puzzles 6 and 7: the 8×8 and “and then?” (page 4)	14
5.4 Puzzles 4 and 5: dominoes (pages 3–4) . .	7	7.4 Puzzles 8 and 9: the 5×5 board (pages 5–6)	15
5.5 Puzzle 6: the flower (page 5, spare)	8	7.5 Puzzle 10: the snake (page 6, spare)	16
6 Table 2–3: dominoes and the checkerboard	9	8 Minutes 52–60: the share	17
6.1 Puzzle 1: how many ways (page 1)	9	9 Background for the adults	18
6.2 Puzzle 2: can it be covered? (page 2) . . .	9		

1 The session at a glance

Minutes	What	Who
0:00–0:05	Free handling of pattern blocks and paper pieces at the tables	every adult at their own table
0:05–0:12	Whole-group launch: the word <i>tiling</i> , the three rules, the mystery board	the organizer, everyone else listens
0:12–0:52	Table work (three separate packets; this guide has a section per table)	one adult per table
0:52–1:00	Share: one discovery per table, then two closing sentences	organizer leads; a child from each table shows

Who sits where

- **K–1 table** (two kindergartners, one first grader): **the parent**. The K–1 pages are read aloud, one box at a time, and Section 5 of this guide is written so that no mathematical background is needed. It is also the table where a calm adult who is good with five-year-olds matters most.
- **Grades 2–3 table** (four third graders): **the second mathematician**. Four children, the most talk, and the checkerboard argument; this table produces the discovery the launch board is about.

- **Grades 4–5 table** (two fourth graders, one fifth grader): **the organizer**. The L-tromino induction and a coloring argument the children have not seen. The organizer steps away for one minute at 0:50 to call the share.

If the parent prefers older children, swap with the 2–3 table; the 2–3 section is also fully scripted. Whoever takes the K–1 table should read Section 5 in full beforehand: the youngest table lives or dies on pacing, not on mathematics.

The idea of the week

Every table meets shapes that can be tiled and shapes that cannot. The mathematics is in the word *cannot*. A tiling can be shown by building it; a non-tiling has to be argued, and the children will find two kinds of argument:

- **a count that every piece respects**. A domino covers two squares, so it can never cover nine. A domino covers one gray and one white square, so a shape with three gray and one white is hopeless even though four is even. A blue rhombus covers one up-pointing and one down-pointing triangle. These are *invariants*: the pieces cannot change them, so the shape must match them.
- **a construction that never stops working**. One tromino in the middle of a 4×4 board turns it into four 2×2 boards with one hole each. The same move turns an 8×8 into four 4×4 s, a 16×16 into four 8×8 s, and so on. That is induction, and the grades 4–5 table will say it in their own words.

What each table brings to the share

K–1: a small triangle that blue rhombi cannot fill, and why. Grades 2–3: the chessboard with two corners missing, and why. Grades 4–5: the doubling trick on the 8×8 board, and why it goes on forever.

2 Materials and preparation

Printing

Print everything at **actual size (100%)**, not “fit to page”: the outlines and boards are drawn at a true 1-inch scale so that the pattern blocks and paper pieces fit them exactly. Check one page with a block before printing the rest.

File	Copies	Notes
week-k-1.pdf	4 (3 children + 1 spare)	page 6 is the domino cut-out: print 2 extra copies of it
week-grades-2-3.pdf	5 (4 children + 1 spare)	page 8 is the domino cut-out: print 2 extra copies of it
week-grades-4-5.pdf	4 (3 children + 1 spare)	page 7 is the tromino cut-out: print 2 extra copies of it
week-facilitator.pdf	3 (one per adult)	the last page is the launch board; print it once more on its own

Cutting (about ten minutes, the evening before)

- **K–1 dominoes**: $2 \text{ sheets} \times 18 = 36$ two-colored dominoes (each is one gray square and one white square). Each child needs about 5.

- **Grades 2–3 dominoes:** 2 sheets $\times$ 18 = 36 plain dominoes. Each child needs up to 7 for the corner-cut boards.
- **Grades 4–5 trominoes:** 2 sheets $\times$ 16 = 32 L-trominoes. Cut each 3×2 block out first, then along the stepped line inside it. Each child needs 5 for a 4×4 board; the 8×8 is drawn, not built.
- Keep each table's pieces in an envelope or cup. Cutting is boring for children and eats the hour; do not plan to have them cut.

Per-table kits

- **K–1:** pattern blocks, the largest share (at least 8 yellow, 16 red, 24 blue, 30 green; the flower on page 5 asks for more, and running out is itself a counting question). The purple chevrons and the nonstandard pieces are fine for free play but set them aside for the counting puzzle on page 1 (see below). The 36 two-colored dominoes. A pencil for the adult.
- **Grades 2–3:** a smaller set of pattern blocks (at least 12 blue and 12 green for page 7), the 36 plain dominoes, pencils, and a dark crayon or colored pencil per child for the checkerboard coloring on page 4.
- **Grades 4–5:** the 32 trominoes, pencils, three colored pencils per child (drawing 21 trominoes on the 8×8 is much easier in alternating colors), about 6 coins or counters to serve as movable holes, and a few sheets of the 1-inch grid paper for building 5×5 boards with the paper trominoes. A few pattern blocks so that the table is not the only one without them.
- **Everyone:** a whiteboard or a large sheet of paper that all three tables can see for the launch, plus the launch board (last page of this guide) or a hand-drawn copy of it.

Two things to know about the pattern blocks

The pieces are related by area: a red trapezoid is three green triangles, a blue rhombus is two, a yellow hexagon is six, and a purple chevron is two blue rhombi, so four triangles. Every counting fact in this packet is a fact about triangles. The children do not need to be told this; it is what they will discover.

Second, the pages name the blocks by color ("blue rhombus", "red trapezoid"), because that is what the children call them. If your set is not in the standard colors, say the shape name instead.

3 Minutes 0–5: free handling

The children arrive, sit at their tables, and get the blocks. Adults sit down too. Do not teach. Ask what they built. If a child covers a region completely, notice it out loud ("no gaps anywhere"); if a child stacks blocks into a tower, that is fine as well. At 0:04, put the envelope of paper pieces on the table without comment. At 0:05 the organizer calls everyone's attention.

4 Minutes 5–12: the launch (organizer)

Seven minutes, five beats. The children stay at their tables and look at the board.

Beat 1 (1 min).

SAY What did you make? Tell me one thing.

Take two or three answers. If anyone covered a region completely, use that as the segue.

Beat 2 (2 min). Draw a hexagon on the board. Draw the line that splits it into two trapezoids.

SAY Today's word is tiling. A tiling covers a shape completely with pieces. Three rules: no gaps, no

overlaps, nothing sticking out past the edge. Say them with me.

Draw the three non-examples quickly: a hexagon with a gap in the middle, two pieces on top of each other, a piece poking outside. Cross each out.

Beat 3 (1 min). Draw a 2×2 square and a domino next to it.

SAY This is a domino: two squares. Can I tile the square with dominoes?

Someone will say two vertical; draw it; someone else will say two horizontal; draw it.

SAY Two ways. Good.

Beat 4 (2 min). Hold up the launch board: the chessboard with two opposite corners cut off.

SAY This chessboard lost two corners. It has 62 squares. Can it be tiled by dominoes? Don't answer, and don't try yet. Every table has its own puzzles today, and at every table you will find shapes that CAN be tiled and shapes that CAN'T, no matter how hard you try. Anybody can say "I couldn't do it." The question we care about is: how could you be SURE that nobody could? At the end, each table shows the rest of us one thing they are sure about.

Beat 5 (1 min). Send them off.

SAY K-1 table: blocks first, then dominoes. Grades 2-3: dominoes, and this board is yours to crack. Grades 4-5: a new piece, the L-tromino, and a board that doubles.

Do not say the words *even*, *odd*, *color*, or *checkerboard* during the launch; those are the tables' discoveries.

5 Table K-1: script for the parent

How to run this table

Three children, one of whom reads a little. You read each shaded box aloud, once, slowly, and then let them work. Your job is to keep hands on blocks, to ask "how do you know?", and to say "I wonder why" instead of explaining. When a puzzle has an impossible part (blue rhombi on the triangle, dominoes on the 3×3), the frustration is the point: say "you are right, it doesn't work, and there is a reason" and go to the shaded picture that follows.

The children do not write. The lines on the pages are for you. Every fact you might need is below; you will not need most of it.

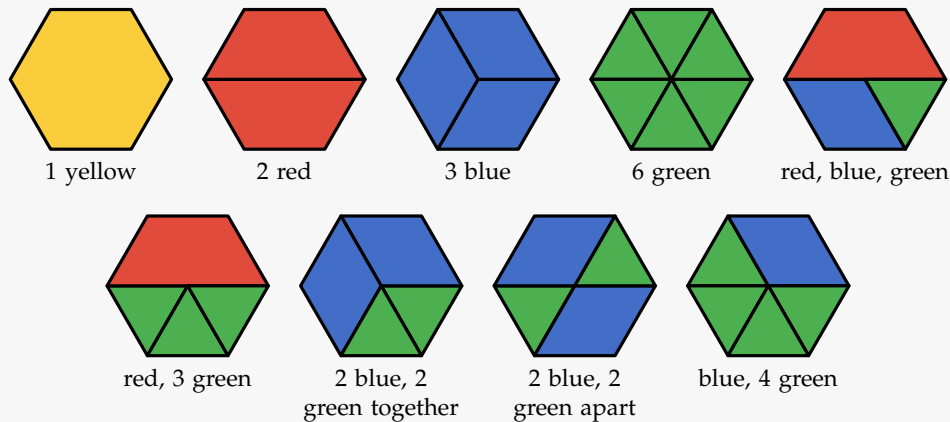
0:12-0:24	Puzzle 1, fill the hexagon	page 1
0:24-0:33	Puzzle 2, the small triangle (the key idea of the table)	page 2
0:33-0:39	Puzzle 3, the big triangle	page 3
0:39-0:50	Puzzles 4 and 5, dominoes	pages 3-4
spare	Puzzle 6, the flower	page 5
0:50-0:52	choose what to show at the share	

5.1 Puzzle 1: fill the hexagon (page 1)

Each child fills hexagons on their own page; when a page is full, they may use a neighbor's. Leave the blocks in place so that the ways can be compared. Set the purple chevrons and the odd pieces aside for this puzzle (they add more ways and muddle the count).

ANSWERS AND ARGUMENTS

There are nine different ways to fill the hexagon with green, blue, red and yellow blocks, counting the plain yellow hexagon as one way (eight if you do not):



Two fillings count as the same way if one is the other turned around or flipped over. Children usually miss the difference between the two “2 blue, 2 green” fillings (greens next to each other, or greens opposite each other): those are genuinely different. Anything from four to nine ways is a fine outcome; the mathematics is in the question “is this one really different from that one?”, which you settle by turning the paper.

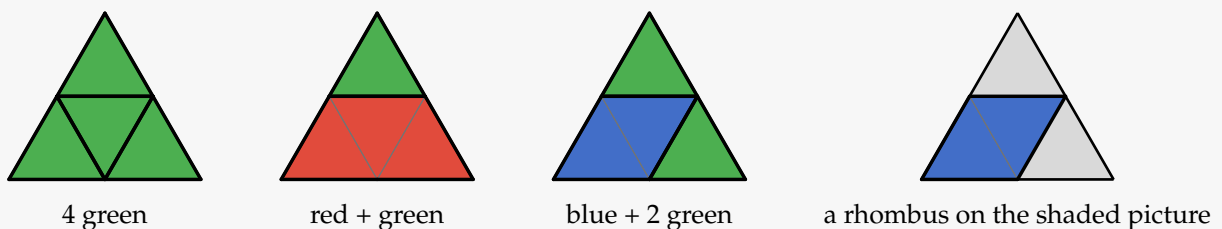
HINTS, IN ORDER

1. SAY *Can you make one using only red? Only blue? Only green?*
2. SAY *Can you make one with exactly three blocks? Exactly four? (Three: 3 blue, or red+blue+green. Four: red+3 green, or 2 blue+2 green.)*
3. SAY *Look at these two. Are they really different, or is one just turned around?*
4. For a quick table:
 SAY *Could you fill it with seven blocks?*
 No: the smallest block is the green triangle, six of them already fill the hexagon exactly, and seven blocks would need more room than there is. That is the first “never” of the day.

5.2 Puzzle 2: the small triangle (page 2)

This is the heart of the K–1 table. Read the first box; let them do the three outlines in order. Green only takes four. Red plus one more: the trapezoid fills the bottom row and a green triangle sits on top. Blue only: it will not work, and they will keep trying; let them try for a couple of minutes.

ANSWERS AND ARGUMENTS



Blue only is impossible. The triangle has three little triangles pointing up (shaded on page 2) and

one pointing down. A rhombus is two little triangles that share a side, and two shaded triangles never share a side (they only touch at corners). So every blue rhombus covers exactly one shaded and one white triangle. Two rhombi would need two white triangles; there is only one. After one rhombus the white triangle is used up and the two remaining shaded triangles cannot be covered by anything blue.

The same fact in domino language: the shaded triangles are the gray squares, the white triangle is the white square, and a blue rhombus is a domino. This is exactly what the other two tables are discovering with checkerboards.

Run the second box as a conversation, not a lecture:

SAY Put one blue rhombus anywhere on the shaded picture. What is under it?

(One shaded, one white.)

SAY Move it somewhere else. What is under it now?

(Still one shaded, one white.)

SAY Can you make it cover two shaded ones?

(They will try; it cannot, because shaded triangles only touch at corners.)

SAY So every blue block uses up one white triangle. How many white triangles are there? So how many blue blocks can we fit at most? And how much is left over?

A kindergartner may not follow all of it; a first grader usually does. Even a child who only takes away “blue never works on this triangle, and it has something to do with the pointing” has what they need for the share.

HINTS, IN ORDER

1. If a child places a rhombus sticking out over the edge:

SAY Remember the third rule.

2. If they insist blue can work:

SAY Show me.

Then count together what is covered and what is not.

3. If the shaded/white explanation does not land, drop it and say:

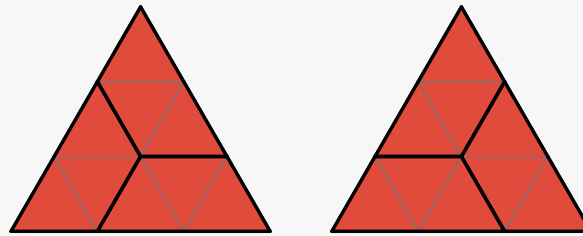
SAY Let's remember this one for the end: blue can't fill it, and the picture with the shading is the reason.

The domino puzzles will bring the idea back.

5.3 Puzzle 3: the big triangle (page 3)

ANSWERS AND ARGUMENTS

Nine little triangles: six pointing up, three pointing down. Three red trapezoids fill it in a pinwheel; there are exactly two ways, mirror images of each other, and both are hard to find by accident, because the obvious start (a trapezoid flat along the bottom) leaves a shape the other two cannot fill.



Blue only is impossible, for either of two reasons, and the children may find either. (1) Nine little triangles; each rhombus covers two; counting by twos never lands on nine. (2) Six point up, three point down; each rhombus covers one of each; after three rhombi the down-pointing ones are gone and three up-pointing ones are left.

HINTS, IN ORDER

1. For the pinwheel:

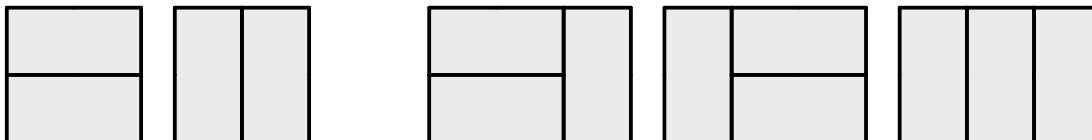
SAY Try putting a red one along each edge, all going the same way around, like a pinwheel.

2. For blue:

SAY Count by twos with me: 2, 4, 6, 8, 10. Is 9 in there?

5.4 Puzzles 4 and 5: dominoes (pages 3–4)

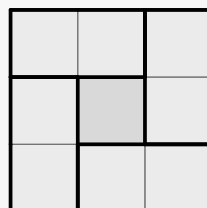
The paper dominoes are one gray square and one white square, like the boards. The 2×2 takes two dominoes (two ways: both standing, or both lying); the 2×3 takes three (three ways).



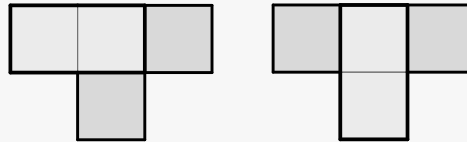
Page 4 has four shapes. Let them try all four before you say anything.

ANSWERS AND ARGUMENTS

- **3×3 square:** no. Nine squares. Every domino covers two, so any number of dominoes covers 2, 4, 6, 8, ..., never 9. Four dominoes fit, and one square is always left over, wherever you put them.



- **Plus sign:** no. Five squares, and five is not in the counting-by-twos list either.
- **Three-square L:** no. Three squares.
- **T shape:** no, and this is the surprise: it has four squares, so two dominoes should be enough, but they never fit. Look at the colors: three gray squares and one white. Each domino covers one gray and one white. Two dominoes would need two white squares; there is only one. After one domino, the leftover two squares are both gray, and they do not touch.



The T is the same argument as the small triangle: gray is “pointing up”, white is “pointing down”, and the domino is the rhombus. If a child says “it’s like the triangle!” the table has done its job.

HINTS, IN ORDER

1. SAY *How many squares does one domino cover? Two dominoes? Three?*
2. For the T:
SAY *Count the gray squares. Count the white ones. What does every domino need?*
3. SAY *Hold your domino up. It is one gray and one white. Can it ever be two grays?*

5.5 Puzzle 6: the flower (page 5, spare)

Seven hexagons. Yellow only: 7. Red only: 14. Blue only: 21. Green only: 42 (you will run out; say so, and count what one hexagon needs and go up by sixes). Any mix works, and “how many blocks did you use” is a different number for every child, which is worth noticing: the flower is the same size, so why the different counts?

If the table stalls

- A child who just builds: let them, for a minute, then hand them one outline and one question (“only red?”). One question at a time.
- A child stuck on an impossible task and getting upset:
SAY *You are right. It can’t be done. You found something true.*
Then the shaded picture.
- Two children arguing about whether two hexagon fillings are the same: turn one page around. Whatever they decide, ask them to stick to it.
- Nobody wants the dominoes: skip to page 4 (the shapes), which is more of a puzzle than page 3.

If the table finishes early

- The flower (page 5).
- SAY *Make me a shape out of green triangles that blue rhombi can never fill. Any odd number of triangles works, and a child who says “I’ll use five” has understood the counting argument.*
- On the 2×3 board:
SAY *How many different ways can you cover it?*
(Three.) Then on the square-grid paper draw a 2×4 : five ways.
- SAY *Can you fill a hexagon with blocks so that no two touching blocks are the same color? (Red, blue, green does it.)*

At 0:50: prepare the share

Pick one child to hold up page 2 with one blue rhombus on the shaded triangle. The sentence to aim for, in the child’s words: “A blue block always covers one shaded and one white. There is only one white. So blue can’t fill it.” You may say it with them.

6 Table 2–3: dominoes and the checkerboard

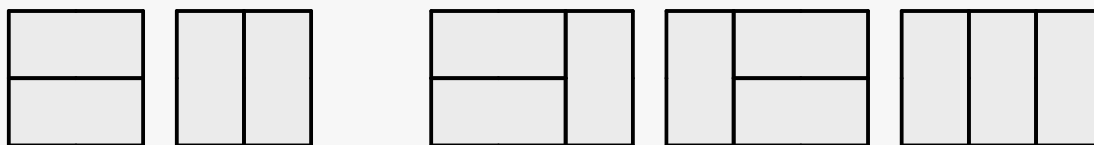
Four third graders who read, write, and argue. Let them work in pairs on the boards and compare as a table. The packet is sequenced so that the coloring is discovered, not announced: pages 2 and 3 produce two impossible shapes with an even number of squares (the T and Board E), and page 4 only then hands over the coloring.

0:12–0:17	Puzzle 1, how many ways (2×2 and 2×3)	page 1
0:17–0:27	Puzzle 2, can it be covered? Rule 1	page 2
0:27–0:34	Puzzle 3, two boards with 14 squares	page 3
0:34–0:43	Puzzle 4, the secret coloring, Rule 2	page 4
0:43–0:50	Puzzles 5 and 6, the big board	page 5
spare	Puzzles 7 and 8	pages 6–7
0:50–0:52	choose what to show at the share	

6.1 Puzzle 1: how many ways (page 1)

ANSWERS AND ARGUMENTS

2×2 : two ways. 2×3 : three ways.



The “same or different” question: the two 2×2 tilings are rotations of each other. We count tilings of a board that stays put, so they are different, and that is the convention the table on page 6 uses (1, 2, 3, 5, 8, 13). If the table decides otherwise, that is fine for pages 1–5; just switch back before page 6.

6.2 Puzzle 2: can it be covered? (page 2)

ANSWERS AND ARGUMENTS

- A, the 3×3 : no. Nine squares; dominoes cover an even number.
- B, the plus: no. Five squares.
- C, the S shape: yes, two horizontal dominoes. It has four squares, like D.
- D, the T: no. Four squares, but no tiling exists. Rule 1 does not explain this one; page 4 will. Let the puzzlement stand:

SAY Put it aside. Something else is going on with the T, and we will come back to it.

Rule 1. A shape with an odd number of squares can never be covered by dominoes, because each domino covers exactly two squares, so any number of dominoes covers an even number of squares.

HINTS, IN ORDER

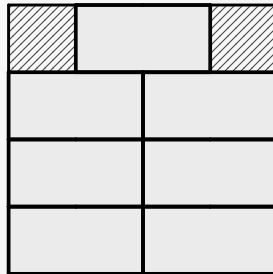
1. *SAY How many squares does one domino cover? Two dominoes? Ten dominoes? Could any number of dominoes cover exactly nine squares?*
2. *SAY Write the number of squares next to each shape. What do the “no” shapes have in common? (They*

will say odd. Then point at the T.)

6.3 Puzzle 3: two boards with 14 squares (page 3)

ANSWERS AND ARGUMENTS

Board E (opposite corners removed): no tiling. Board F (two corners on the same side removed): tileable, for example



The point of the page is the question at the bottom: how could you be *sure* that E is impossible without trying every way? Give them a few minutes with it before turning to page 4. If someone says “color it like a chessboard” or “the corners are the same color”, hand them the crayons and let them lead.

6.4 Puzzle 4: the secret coloring (page 4)

ANSWERS AND ARGUMENTS

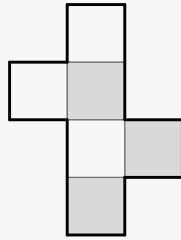
Whichever square a child starts with, squares sharing a side get opposite colors, so the two counts are determined up to swapping. Using the shading of page 5 (corner squares gray):

shape	shaded	white	covered?
D (T)	3	1	no
A (3×3)	5	4	no
B (plus)	1	4	no
C (S)	2	2	yes
E (opposite corners)	6	8	no
F (same-side corners)	7	7	yes

A domino always covers one shaded and one white square, because the two squares it covers share a side. So 7 dominoes cover 7 shaded and 7 white. **Rule 2:** a shape can only be covered by dominoes if it has exactly as many shaded squares as white squares. The T has 3 and 1; Board E has 6 and 8 (the two removed corners were both gray). Neither can be covered, and nobody had to try every way.

Rule 2 swallows Rule 1: an odd number of squares cannot split into two equal halves.

For you, not necessarily for the children: Rule 2 is necessary but not sufficient. The six-square shape below has three shaded and three white squares and still has no tiling, because the top square and the left square each have only one neighbor, and it is the same neighbor. The full condition is Hall’s marriage theorem (Section 9).

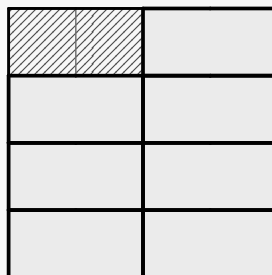
**HINTS, IN ORDER**

1. SAY *Put a domino anywhere on your colored T. What is under it? Move it. Now?*
2. SAY *If two dominoes could cover the T, how many shaded squares would they cover altogether?*
3. SAY *What did the two missing corners of Board E have in common?*

6.5 Puzzles 5 and 6: the big board (page 5)**ANSWERS AND ARGUMENTS**

The 8×8 board has 32 gray and 32 white squares (every row has four of each). Opposite corners have the same color: on page 5, extending the pattern into the two missing corners makes both of them gray. Removing them leaves 30 gray and 32 white. Thirty-one dominoes would cover 31 of each. So no tiling exists. Fast counting: four per row, eight rows, then subtract the two corners from the gray count.

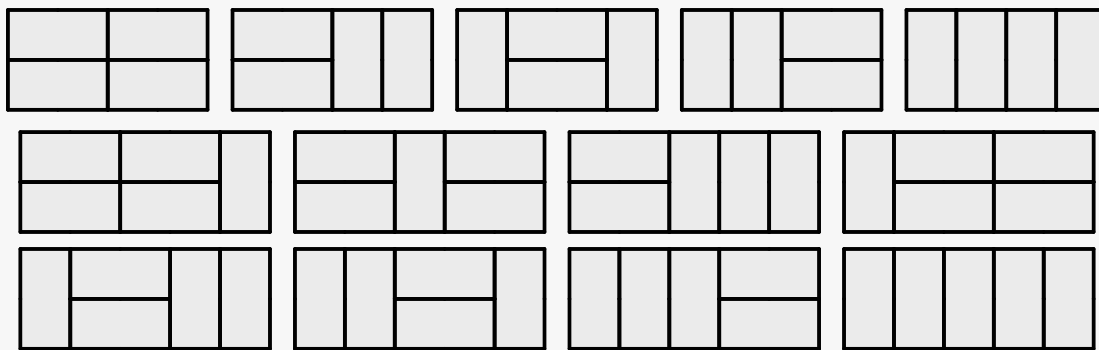
Puzzle 6: the 4×4 board with two neighboring squares of different colors removed can be covered, for example



and the general answer is yes: any board with one square of each color removed, wherever they are, can be tiled (Gomory's theorem). The 4–5 table has the proof as a challenge (Section 7, Puzzle 10); the idea is a closed path through all 64 squares. Tell the children it is true and that it is a good question to ask the older table at the share.

6.6 Puzzles 7 and 8: extensions (pages 6–7)**ANSWERS AND ARGUMENTS**

$2 \times n$ boards. 2×4 : 5 ways. 2×5 : 8 ways. 2×6 : 13 ways. Each number is the sum of the previous two. Reason: look at the left end of a $2 \times n$ board. Either a vertical domino stands there, leaving a $2 \times (n - 1)$ board, or two horizontal dominoes lie there one above the other, leaving a $2 \times (n - 2)$ board; nothing else can cover the top-left square without forcing one of those two. So $\text{ways}(n) = \text{ways}(n - 1) + \text{ways}(n - 2)$. These are the Fibonacci numbers.



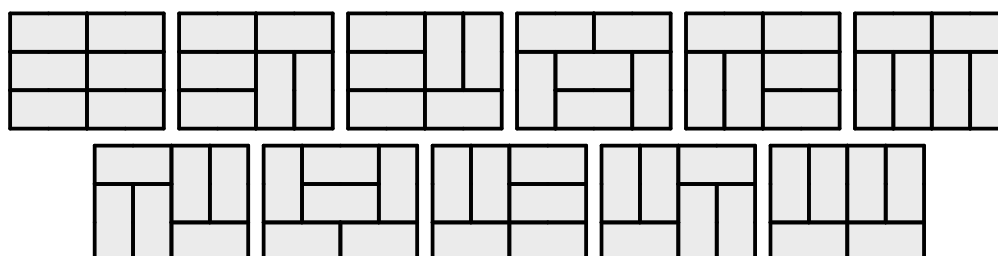
The side-4 triangle. Sixteen little triangles, ten pointing up and six pointing down. A blue rhombus covers one of each, so blue rhombi alone can never fill it, even though 16 is even. No triangle of this kind can be filled with rhombi: a triangle of side n has n more up-pointing than down-pointing little triangles.

If the table stalls

- On Puzzle 2, if nobody sees “odd”: have them write the number of squares under each shape and sort the shapes into two piles.
- On Puzzle 3, if they keep trying Board E:
 SAY *You have tried ten ways. There are thousands. Is there a way to know without trying them all?*
 Then page 4.
- On Puzzle 4, if the coloring is done but nobody looks at a domino:
 SAY *Pick up one domino and put it on the colored board. What is under it? Does it matter where you put it?*
- If the writing slows everything: let one child be the scribe and the others talk.

If the table finishes early

- Pages 6 and 7, in that order.
- SAY *Draw a shape with the same number of shaded and white squares that still cannot be covered. The six-square shape above is the smallest connected one; a shape made of two separate squares of different colors is the cheap answer, and it is a good answer.*
- On square-grid paper: how many ways to cover a 3×4 board? Eleven; finding all eleven as a table is a solid ten minutes.



- SAY *Can you tile the 8×8 board with two squares of different colors removed? Any pair they choose works; let them find the tiling.*

At 0:50: prepare the share

One child holds up the colored Board E (page 4) or the big board (page 5). The sentence: “Every domino covers one gray and one white. The two corners we lost were both gray. So there are two more white than gray, and it can’t be done.” A second child can add “and we didn’t have to try every way.”

7 Table 4–5: the L-tromino and the board that doubles

Three children who like a challenge. The packet is one long argument: a 4×4 with any hole can be tiled (by trying), a trick explains why (one tromino in the middle), the trick doubles the board (the 8×8), and the trick never stops (induction). Then a board where counting says maybe and a coloring says no (the 5×5), and a challenge in the other direction (the snake).

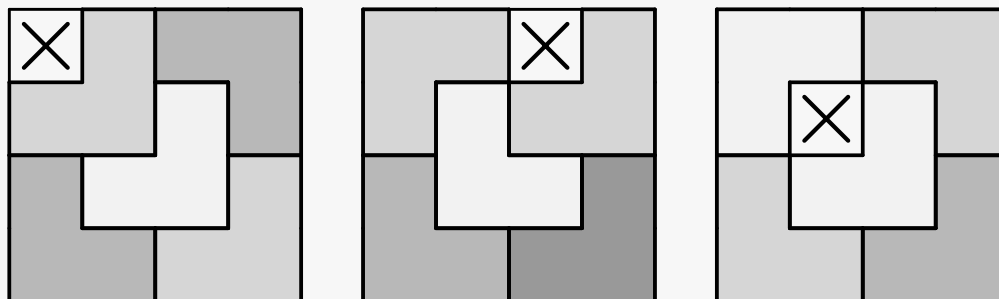
0:12–0:15	Puzzles 1 and 2, warm-up and the 4×4	page 1
0:15–0:23	Puzzle 3, move the hole	page 2
0:23–0:31	Puzzles 4 and 5, the trick and the counting check	page 3
0:31–0:41	Puzzles 6 and 7, the 8×8 and “and then?”	page 4
0:41–0:50	Puzzles 8 and 9, the 5×5	pages 5–6
spare	Puzzle 10, the snake	page 6
0:50–0:52	choose what to show at the share	

7.1 Puzzles 1–3: the 4×4 with a hole (pages 1–2)**ANSWERS AND ARGUMENTS**

Puzzle 1: one tromino; a 2×2 board with a hole is a tromino.

Puzzle 2: five trominoes, because 15 squares must be covered and each tromino covers 3; no other number is possible.

Puzzle 3: every position of the hole works. By symmetry there are only three kinds of position (corner, edge, inside), and pages 1–2 have one of each:



Trial and error finds these quickly. Do not give the trick yet; the point of page 3 is that they meet the trick after they have felt the need for a method.

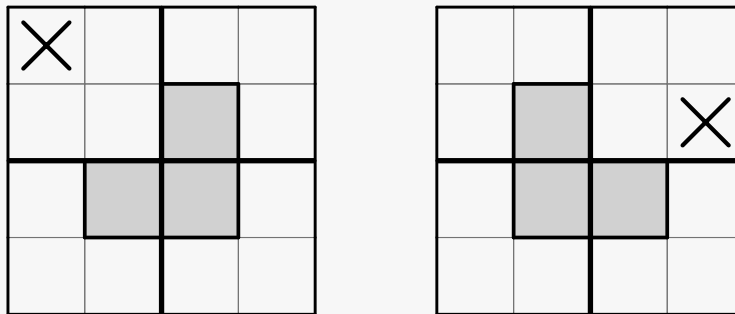
HINTS, IN ORDER

1. SAY *Each corner of the board has to be covered by something. What are the choices for a corner square?*
2. SAY *Which squares are the hardest to cover? Start there.*

7.2 Puzzles 4 and 5: the trick and the counting check (page 3)

ANSWERS AND ARGUMENTS

Cut the board into four 2×2 boards. The hole lies in one of them. The other three meet at the center of the board, and one tromino placed at the center covers exactly one square from each of them:



Now each 2×2 board is missing exactly one square, either the original hole or a square under the central tromino, and a 2×2 board missing one square is exactly one tromino. Four more trominoes finish it. This works for every position of the hole, because whichever quadrant the hole is in, the central tromino is placed in the other three.

Counting check: 4×4 : 15 and 5. 8×8 : 63 and 21. 16×16 : 255 and 85. 32×32 : 1023 and 341. Every entry in the middle column is a multiple of 3. The children can see it from the numbers; the tiling is a *proof* of it: a $2^n \times 2^n$ board minus one square is tiled by trominoes, so its number of squares, $4^n - 1$, is a multiple of 3. No division required.

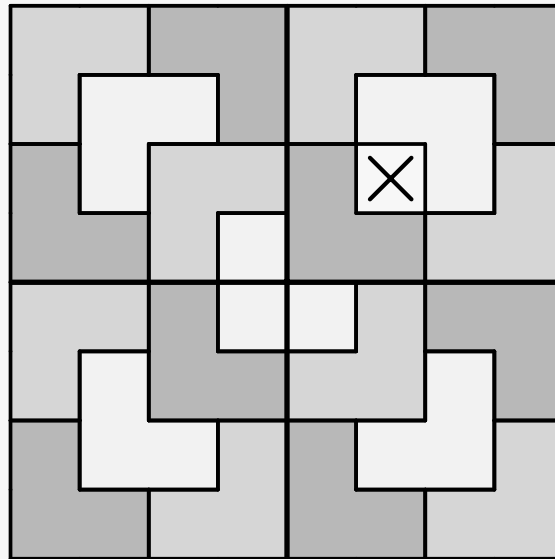
HINTS, IN ORDER

1. SAY *The hole is in one of the four small boards. What do the other three small boards need? (Each needs to lose one square.)*
2. SAY *Is there one tromino that touches three of the small boards at once?*

7.3 Puzzles 6 and 7: the 8×8 and “and then?” (page 4)

ANSWERS AND ARGUMENTS

Cut the 8×8 into four 4×4 boards. One tromino at the center covers one square from each of the three 4×4 s that do not contain the hole. Now each 4×4 has exactly one missing square, and page 3 showed how to finish any 4×4 with any hole. Twenty-one trominoes altogether. A complete example, hole in the upper right quadrant:



The children should stop drawing once the method is clear; a finished board is nice but not necessary.

“And then?”: the 16×16 . Cut it into four 8×8 s, one tromino at the center, and each 8×8 now has one hole, which we can do. The 32×32 the same way, using the 16×16 . It never stops: *if we can do a board of some size with any hole, we can do the board of double the size with any hole*, and we can do the 2×2 , so we can do 4, 8, 16, 32, ..., forever.

This is the shape of an argument mathematicians call *induction*: a first rung, and a rule for climbing from any rung to the next. Say the word once, and let the children’s own sentence stand as the definition.

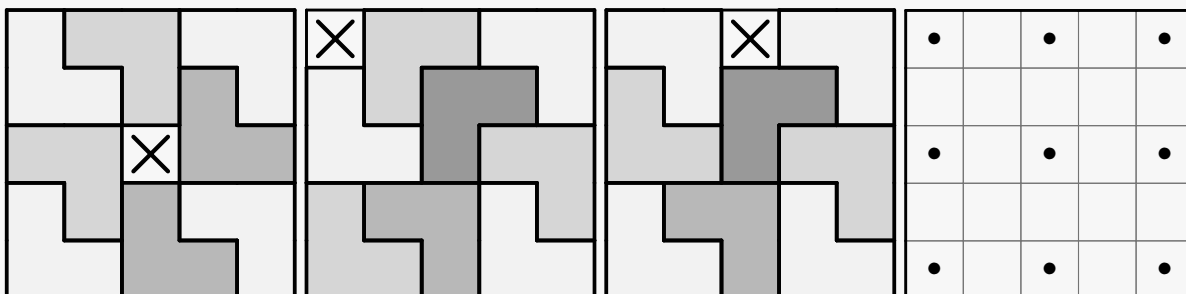
HINTS, IN ORDER

1. SAY Where are the four 4×4 boards? Draw the two lines.
2. SAY After the middle tromino, what does each 4×4 board look like? Have you seen that before?
3. If the drawing gets messy: alternate three colors, or write a number in each tromino.

7.4 Puzzles 8 and 9: the 5×5 board (pages 5–6)

ANSWERS AND ARGUMENTS

A 5×5 with one hole has 24 squares, so counting allows 8 trominoes. But it depends on the hole. Exactly nine positions work: the squares in an odd row and an odd column (corners, midpoints of the edges, and the center). The other sixteen do not.



Board G (hole next to a corner) is impossible, and the children will feel it after a few tries. Page

6 gives the reason, step by step: color the nine odd-odd squares black. Every tromino lies inside some 2×2 box (it is a 2×2 box minus one square). A 2×2 box contains exactly one black square, because among any two consecutive rows exactly one is odd, and the same for columns. So a tromino covers at most one black square; eight trominoes cover at most eight. If the hole is not black, all nine black squares must be covered. Nine is more than eight. So the hole must be black. This is the same kind of argument as the 2–3 table’s checkerboard, with “at most one” in place of “exactly one”. The coloring shows only that the other sixteen holes fail; that the nine black holes succeed is shown by the three tilings above and symmetry.

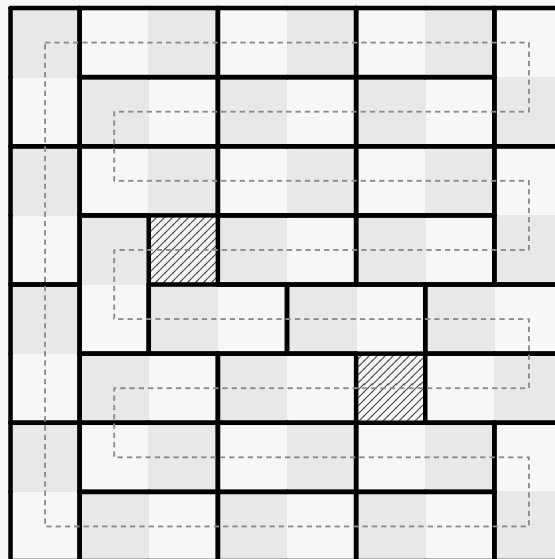
HINTS, IN ORDER

1. SAY Which holes worked? Mark them on the summary grid. What pattern do you see?
2. SAY Every tromino fits in a 2×2 box. Color the board so that every 2×2 box has exactly one colored square.
3. Page 6, questions (a)–(e), in order. Answers: 1; 1; 8; 9; the hole must be one of the nine black squares.

7.5 Puzzle 10: the snake (page 6, spare)

ANSWERS AND ARGUMENTS

The snake is a closed path through all 64 squares, each step moving to a neighboring square, so the colors along the path alternate gray, white, gray, white, ... Remove one gray square and one white square. The loop breaks into two paths (one of them may be empty). Walking along the loop from the gray hole to the white hole, the squares strictly between them alternate starting with white and ending with gray, so there is an even number of them; the same holds for the other path. Lay dominoes along each path, two squares at a time. Done, wherever the two holes were.



This is Gomory’s theorem, and it answers the 2–3 table’s Puzzle 6. It also shows what the coloring cannot: a coloring proves impossibility; a construction proves possibility.

If the table stalls

- Puzzle 3 dragging (they cannot tile some position):

SAY *Which quadrant is the hole in? What is left in the other three?*

That is the trick, delivered as a hint.

- Puzzle 4, the trick does not come: draw the two lines yourself and place the central tromino without a word. Ask what each small board now looks like.
- Puzzle 6, the 8×8 overwhelming:

SAY *Don't finish it. Convince me you could.*

- Puzzle 8, the coloring does not come: it is not supposed to come unaided; page 6 walks them through it.

If the table finishes early

- The snake (Puzzle 10), then:

SAY *Does the snake trick work on a 4×4 ? On a 6×6 ? What does the board need?*

(Any board with a closed path through all squares. Every rectangle with an even number of squares and both sides at least 2 has one.)

- SAY *What about a 7×7 with one hole?* $48 = 16 \times 3$, so counting allows it. Every one of the 49 positions works (Chu and Johnsonbaugh, 1986); let them test the hole next to a corner, which fails on the 5×5 and succeeds here. This is a good “prediction, then test” exercise; do not try to prove it.
- SAY *Which rectangles can trominoes tile with no hole at all?* 2×3 yes, 3×4 yes, 2×4 no (8 is not a multiple of 3), 3×3 no (try it: the center square forces a contradiction), 3×5 no, 5×6 yes. Let them collect data.
- The 2–3 table's $2 \times n$ Fibonacci count (Section 6, Puzzle 7) if they have not seen it.

At 0:50: prepare the share

One child holds up a drawn 8×8 (page 4), finished or not, with the two heavy lines and the central tromino visible. The sentence: “One tromino in the middle turns a big board into four smaller boards with one hole each. So if you can do the small board, you can do the big one. And then the next one. Forever.” If the 5×5 went well, a second child can add: “and on the 5×5 only nine holes work, because of a coloring.”

8 Minutes 52–60: the share

The organizer calls the room together at 0:52. Children stay at their tables and turn their chairs. Order: K–1 first (their attention is shortest), then 2–3, then 4–5. Two minutes each, one minute for the closing.

For each table the adult says one sentence of setup (“Our table tried to fill this triangle with blue rhombi”), the child shows the object and says their sentence, and the organizer asks the room one question:

- after K–1:

SAY *The T-shape at the K–1 table and the small triangle: what do they have in common?*

(Two colors, and every piece takes one of each.)

- after 2–3:

SAY *So is the launch board possible?*

No, and now everyone knows why. Hold it up and count the corners' colors together.

- after 4–5:

SAY *Could the 4–5 table do a board of size 1024? Have they drawn it?*

No, and they do not need to.

Closing (organizer).

SAY Three tables, three arguments. Two were counting arguments: color the shape so that every piece takes the same amount of each color, and if the shape's counts don't match, there is no tiling, and nobody had to try every way. One was a building argument: a trick that turns a big board into small boards, so it works for boards of every size, forever. Mathematicians use both kinds every day. And here is a secret: for a strange shape and a strange piece, nobody in the world knows a quick way to decide whether a tiling exists. That is a real research question, and today you took the first step into it.

Likely questions, with answers: “Is there a shape that fools the coloring?” Yes; the six-square shape in Section 6. “How many ways can you tile the whole chessboard with dominoes?” 12,988,816. “Can you always tile it if you remove one gray and one white square?” Yes, and the 4–5 table may have the snake to show.

9 Background for the adults

None of this is for the children; it is so that the two mathematicians can answer questions honestly and the parent can enjoy the day.

Dominoes and matchings. A domino tiling of a region is a perfect matching in the graph whose vertices are the squares and whose edges join squares that share a side. That graph is bipartite (the two colors), so the coloring count is the bipartite balance condition, and Hall's theorem (1935) gives the exact condition: every set S of gray squares must have at least $|S|$ white neighbors. The six-square shape fails Hall's condition with S the two leaf squares. Matchings can be found in polynomial time, so deciding domino tileability of a finite region is easy; *counting* tilings is a different matter, and Kasteleyn and, independently, Temperley and Fisher (1961) found the exact formula for rectangles, giving 12,988,816 for the 8×8 board and the Fibonacci numbers for $2 \times n$.

Gomory's theorem (any two squares of opposite color can be removed from the chessboard and the rest tiled) is the Hamiltonian-cycle argument of Puzzle 10; it appears in Honsberger's *Mathematical Gems I* (1973).

Trominoes. Golomb introduced polyominoes and proved the $2^n \times 2^n$ theorem in “Checker boards and polyominoes” (*American Mathematical Monthly*, 1954). Chu and Johnsonbaugh (“Tiling deficient boards with trominoes”, *Mathematics Magazine*, 1986) settled the square boards of every size: when n is not a multiple of 3, an $n \times n$ board with one hole can be tiled for every position of the hole, except $n = 5$, where exactly the nine odd-odd positions work. The nine-square coloring on page 6 of the 4–5 packet is the standard obstruction.

Colorings and beyond. Colorings are the first rung of a ladder of invariants. Conway and Lagarias (*J. Combinatorial Theory A*, 1990) showed how to attach a group-theoretic invariant to a tiling problem that decides cases no coloring can settle, for example the triangular regions tiled by three-in-a-row “tribones”; Thurston's “Conway's tiling groups” (*American Mathematical Monthly*, 1990) turns the domino case into height functions and gives a linear-time tiling algorithm for simply connected regions.

Hardness. Deciding whether a finite region can be tiled by L-trominoes is NP-complete (Moore and Robson, “Hard tiling problems with simple tiles”, *Discrete and Computational Geometry*, 2001), so the contrast between the dominoes (easy) and the trominoes (hard) is real and sharp. For tilings of the whole plane by a given finite set of square tiles with colored edges, Berger (1966) proved that no algorithm can decide whether a tiling exists at all. That is the sentence behind the closing line of the share.

The launch board

A chessboard with two opposite corners removed: 62 squares.

